

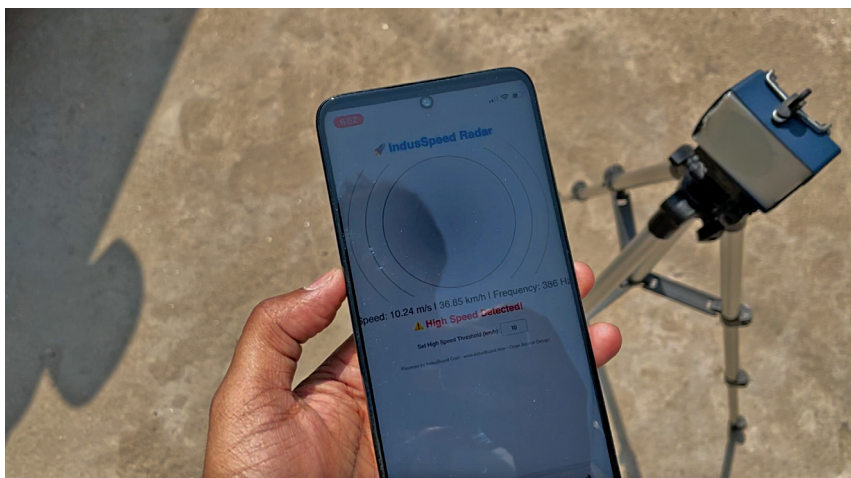


Fig. 6: Radar measuring cars' speed on the road, and alerting in case of over-speeding

the frequency of the returning signal remains mostly unchanged. However, if the object is in motion relative to the radar, the reflected signal undergoes a Doppler shift.

A positive Doppler shift (increased frequency) occurs when an object approaches, while a negative Doppler shift (decreased frequency) happens when an object moves away. By analysing this frequency shift, the system can accurately determine the speed of the moving object in real time, enabling precise detection without physical contact or complex setups.

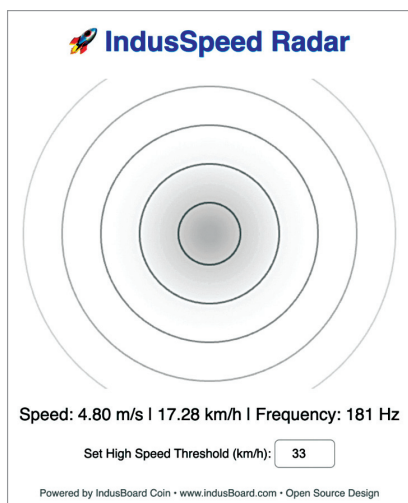


Fig. 7: The web page UI showing the radar waves and data in grey, the frequency of Doppler effect, and speed in m/s and km/h

Circuit and working

Fig. 2 shows the circuit diagram of the automatic Doppler radar system. It is built using the IndusBoard Coin, HB100 Doppler radar module, and LM358 amplifier. This system processes real-time speed data and displays it on a wireless dashboard that is accessible directly from a smartphone, without requiring an internet connection.

The output signal from the HB100 sensor is extremely weak, usually in the range of a few millivolts, and cannot be directly processed by the microcontroller. An LM358 op-amp-based amplifier

module is used to overcome this, which takes the HB100's IF (intermediate frequency) output and boosts it to a detectable voltage level. This amplified signal is then fed into a digital input pin (GPIO4) of the IndusBoard Coin.

The power connections are straightforward. The HB100 and the amplifier module are powered using the 5V and GND pins from the IndusBoard Coin. The output from the amplifier is connected to GPIO4, which is configured as an interrupt pin to allow accurate pulse counting by the microcontroller.

Software

The firmware for the system is written using the Arduino IDE. The code begins by initialising the serial port and configuring GPIO4 as an input pin with an interrupt attached.

The interrupt service routine (ISR) is triggered on every rising edge of the pulse from the radar sensor, incrementing a 'pulseCount' variable. This pulse count represents the number of Doppler shift cycles detected per second.

Every one second, the main loop reads the 'pulseCount', resets it, and uses it to calculate the speed. The formula used to convert pulse



Fig. 8: Radar deployed on a roadside tripod to measure the speed of incoming vehicles